CubeSats, FOSS4G, and cubesatdata.com

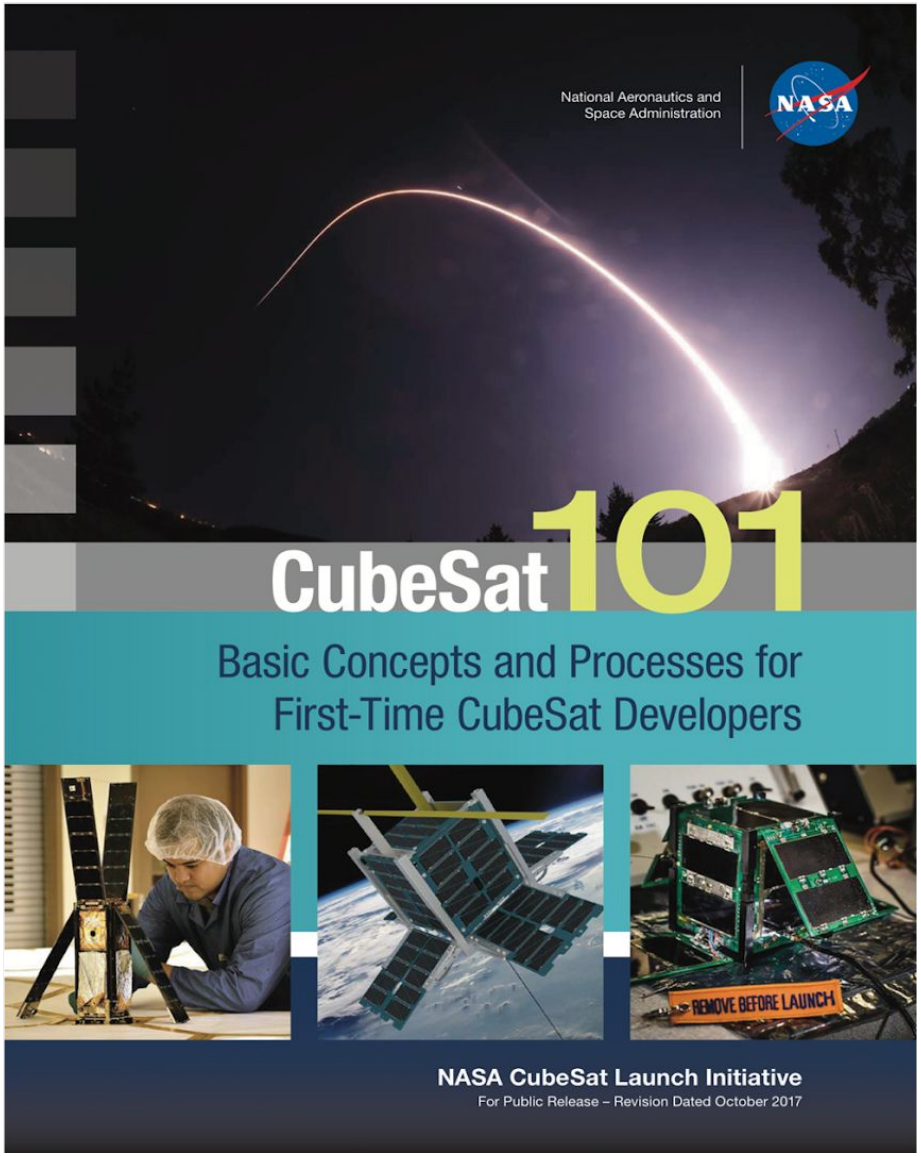
Matthew Hanson

mhanson@element84.com

FOSS4G 2019

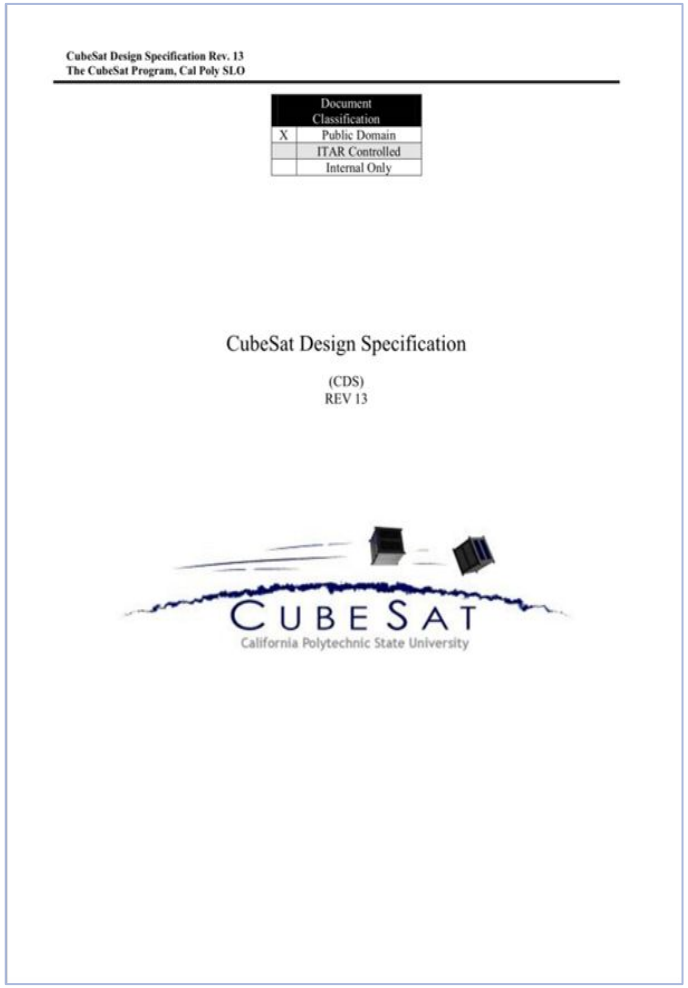
Element 84

CubeSat 101


National Aeronautics and Space Administration
NASA
CubeSat 101
Basic Concepts and Processes for First-Time CubeSat Developers
REMOVE BEFORE LAUNCH
NASA CubeSat Launch Initiative
For Public Release – Revision Dated October 2017

What are CubeSats

Strict specifications allows for mass production of parts, reducing cost.


CubeSat Design Specification Rev. 13
The CubeSat Program, Cal Poly SLO

	Document Classification
X	Public Domain
	ITAR Controlled
	Internal Only

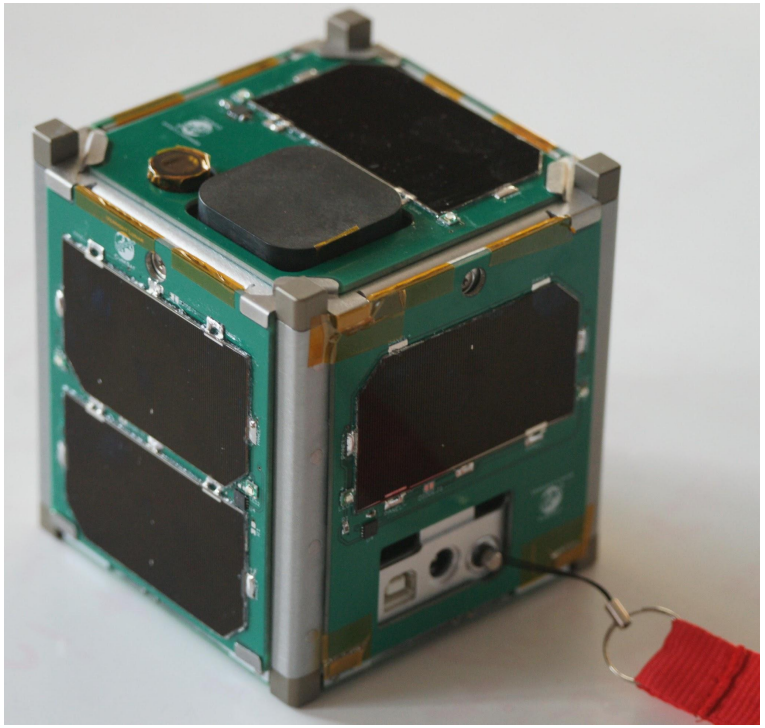
CubeSat Design Specification

(CDS)
REV 13

CUBESAT
California Polytechnic State University

CubeSat Design Specfication (CDS): http://www.cubesat.org

What are CubeSats

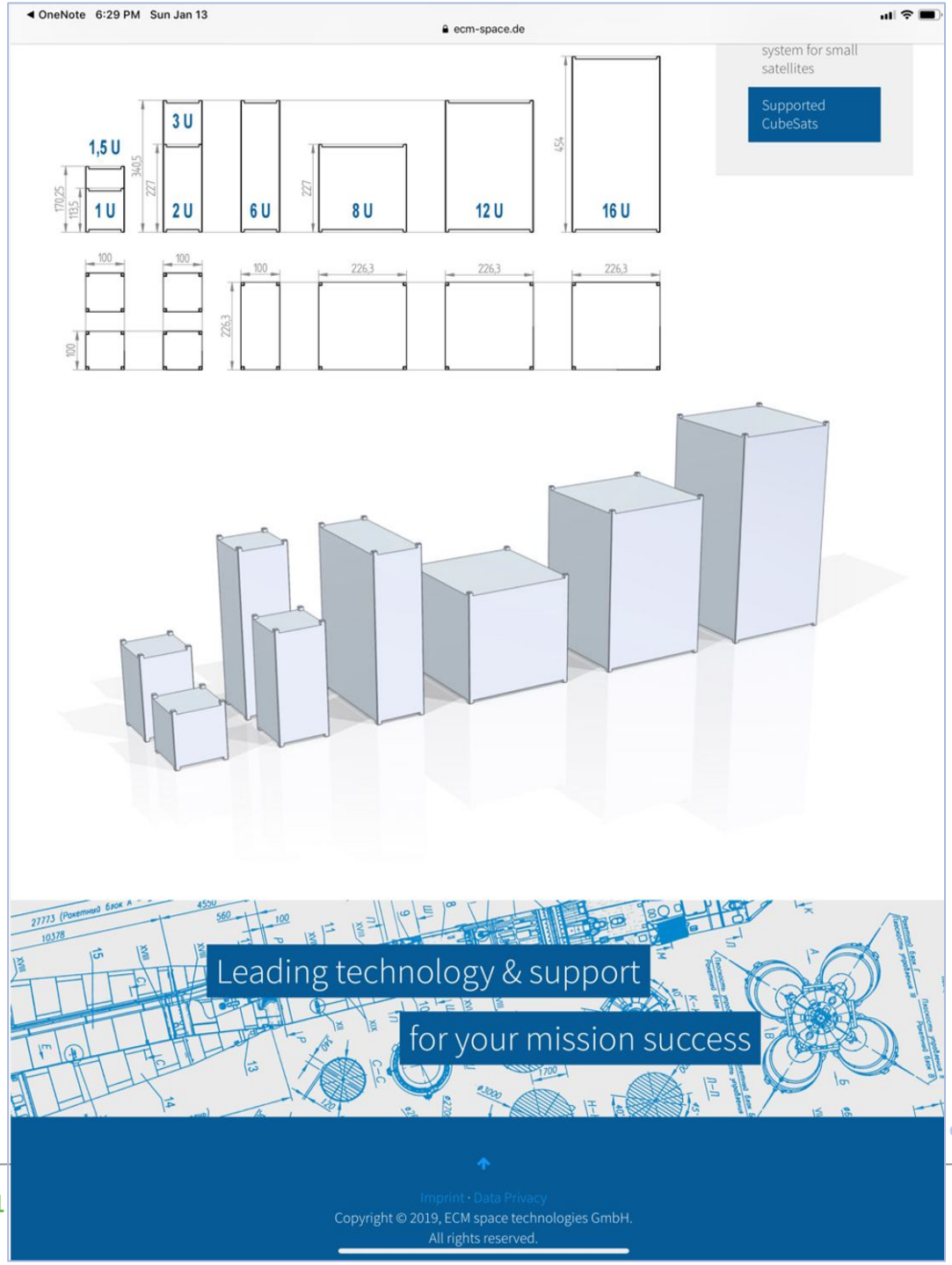


Satellite mass classification

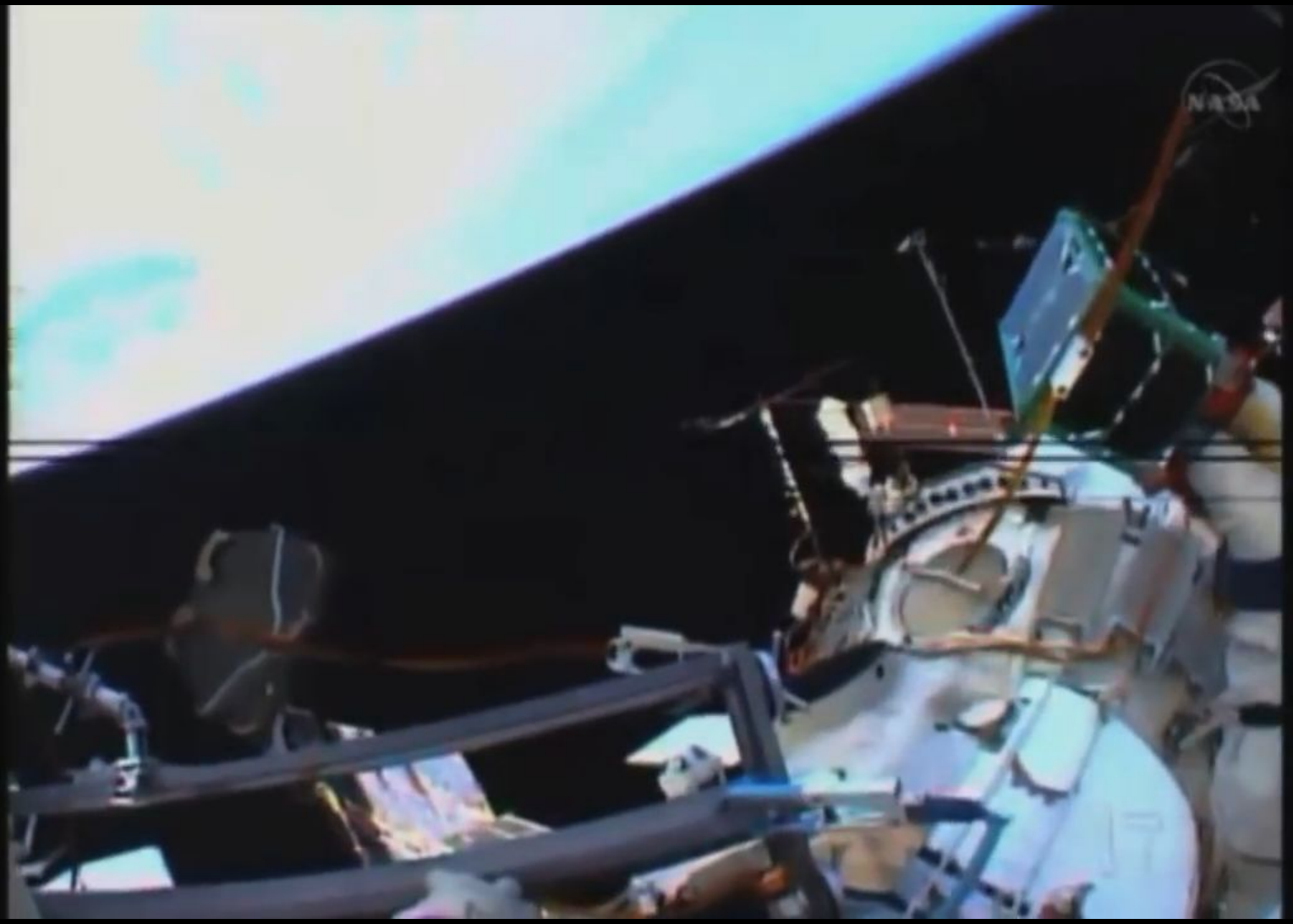
- Large satellites: >1000 kg
- Medium satellites: 500 to 1000 kg
- Small satellites: < 500 kg
  - Minisatellites: 100 to 500 kg
  - Microsatellites: 10 to 100 kg
  - Nanosatellites: 1 to 10 kg
  - Picosatellites: 100 g – 1 kg
  - Femtosatellites: 10 g – 100 g
  - Attosatellites: 1 g – 10 g
  - Zeptosatellites: 0.1 g – 1 g
- CubeSat sizes:
  - From ~0.2 kg to ~40 kg
  - From 0.25U to 27U

By Cbrandonvt - Own work, CC BY 3.0
https://commons.wikimedia.org/w/index.php?curid=37677790

nanosats.eu/cubesat.html



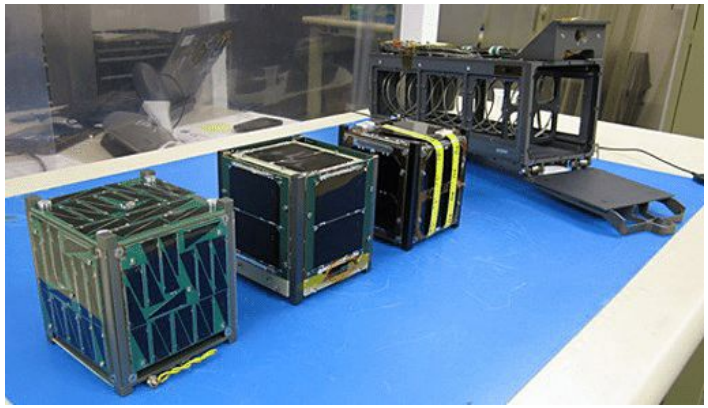
ecm-space.de/index.php/launch-adapters-h/cubesat-sizes

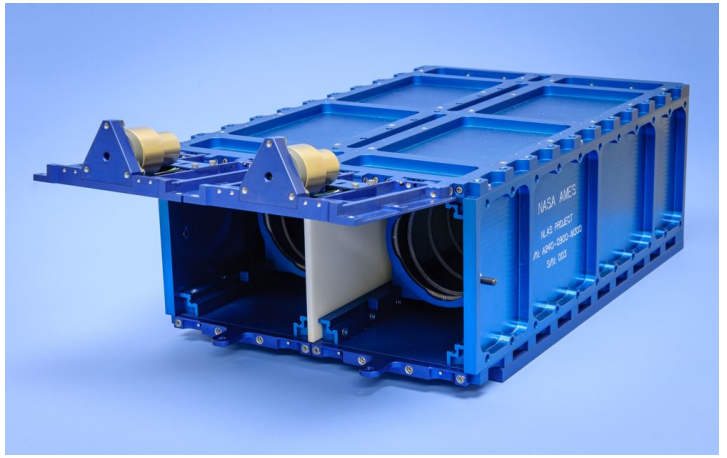

NASA
HD


Port of Baltimore USA
Port of Baltimore USA
Port of Baltimore USA
EVERGREEN
CHINA SHIPPING
CHINA SHIPPING
COSCO
COSCO
COSCO
EVERGREEN
EVERGREEN
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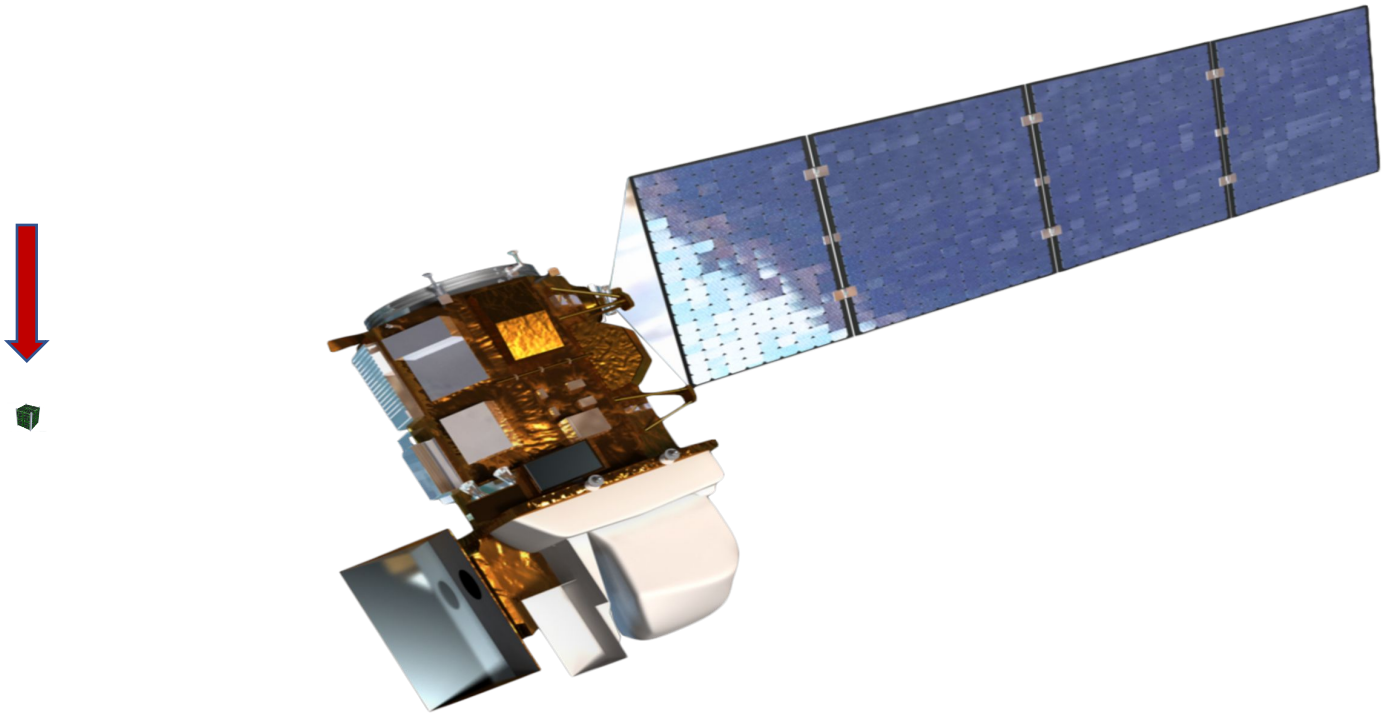
CubeSat Dispenser Systems

- Poly-Picosatellite Orbital Deployer (P-POD) supports up to 3U worth of CubeSats.
- Bigger dispensers now available
- Kind of bolted onto whatever won't fall off.
- Launch vehicle says when to release, door opens, satellites stream out.



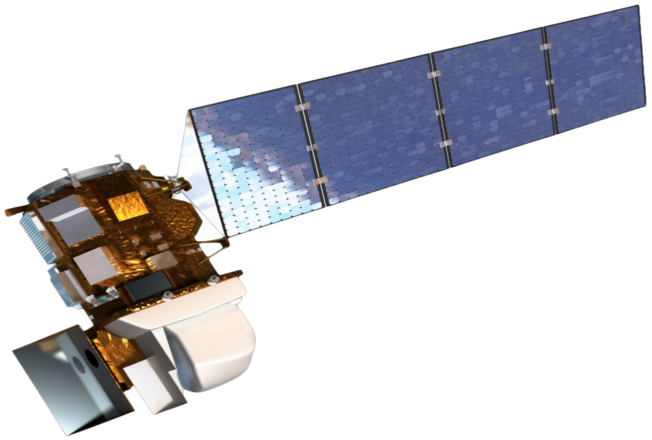


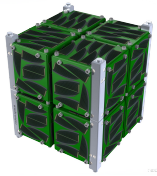
https://www.nasa.gov/centers/ames/engineering/small-sat/nlas
https://www.researchgate.net/figure/Three-1U-CubeSats-beside-a-3U-Poly-Picosatellite-Orbital-Deployer-PPOD-developed-at_fig2_317036984



landsat.gsfc.nasa.gov/landsat-8/landsat-8-overview/
https://www.nanosats.eu/cubesat.html

Different Goals





- Baseline science quality data
- 2623kg
- 3m long, 2.4m diameter
- ~$850 billion
- 5+ year design life (Landsat 5 flew for 29 years)

- Experiments, rapid iteration, “good enough” data
- ~1-1.33kg
- 10cm Cube
- ~$40k + Launch Fees
- Weeks to months design life

landsat.gsfc.nasa.gov/landsat-8/landsat-8-overview/

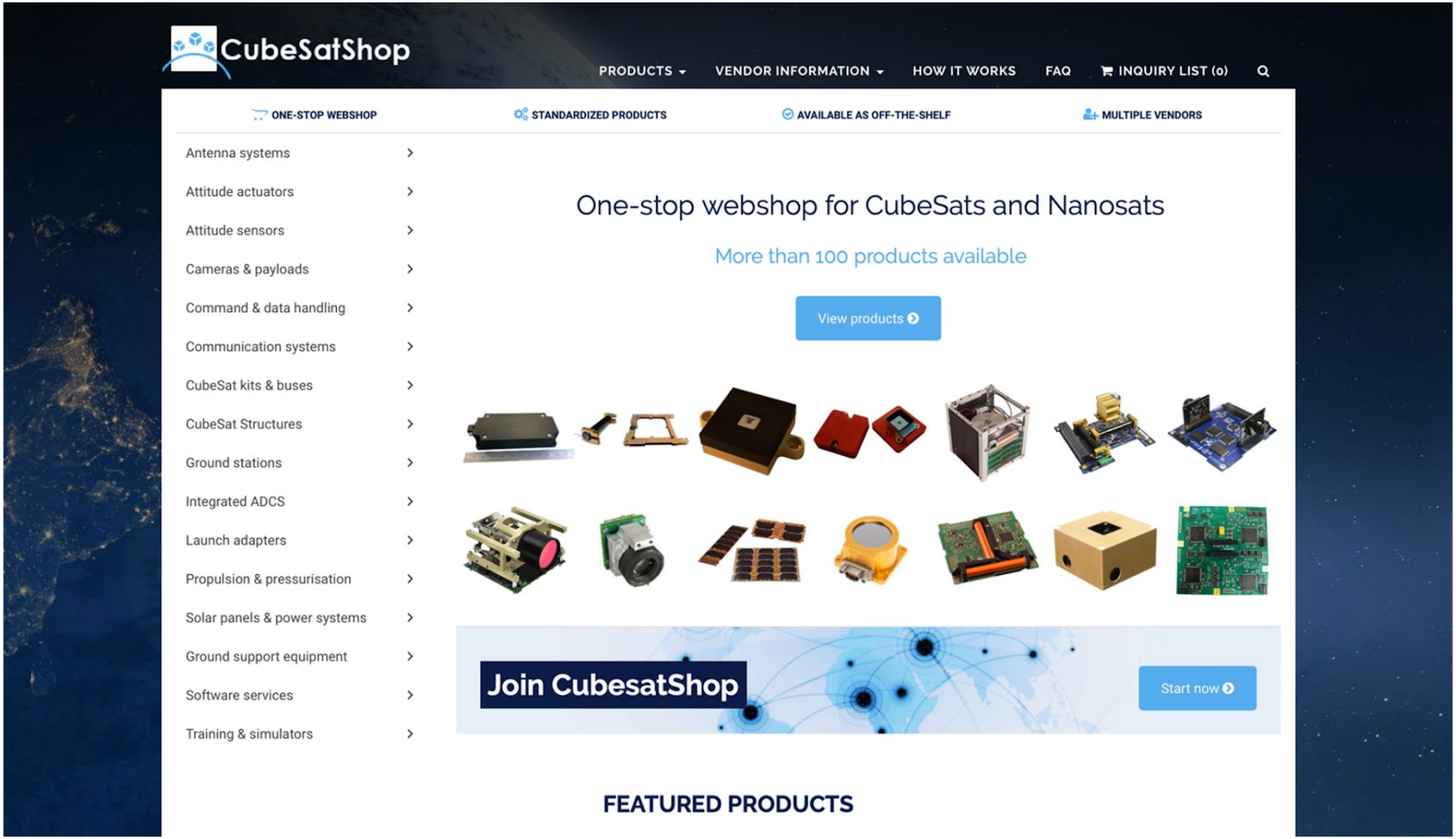
https://www.nanosats.eu/cubesat.html

Analysis Ready Data for CubeSats

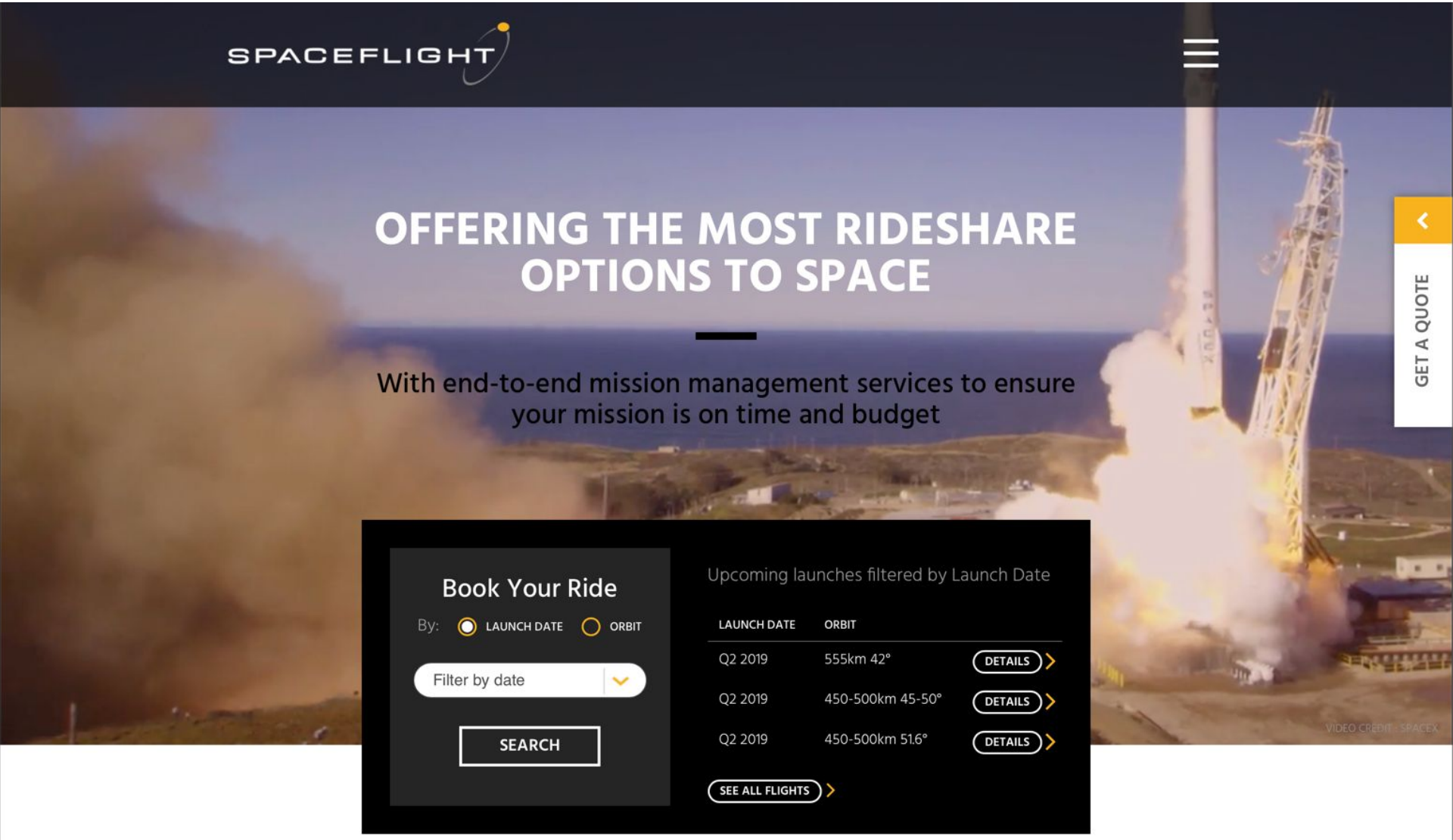
- Cheaper hardware means
  - relying more on software for calibration and correction

- See talk by Chris Rampersad of Urthecast from ARD19
  - “Building Analytics Ready Data for Small Satellites”

cubesatshop.com



spaceflight.com

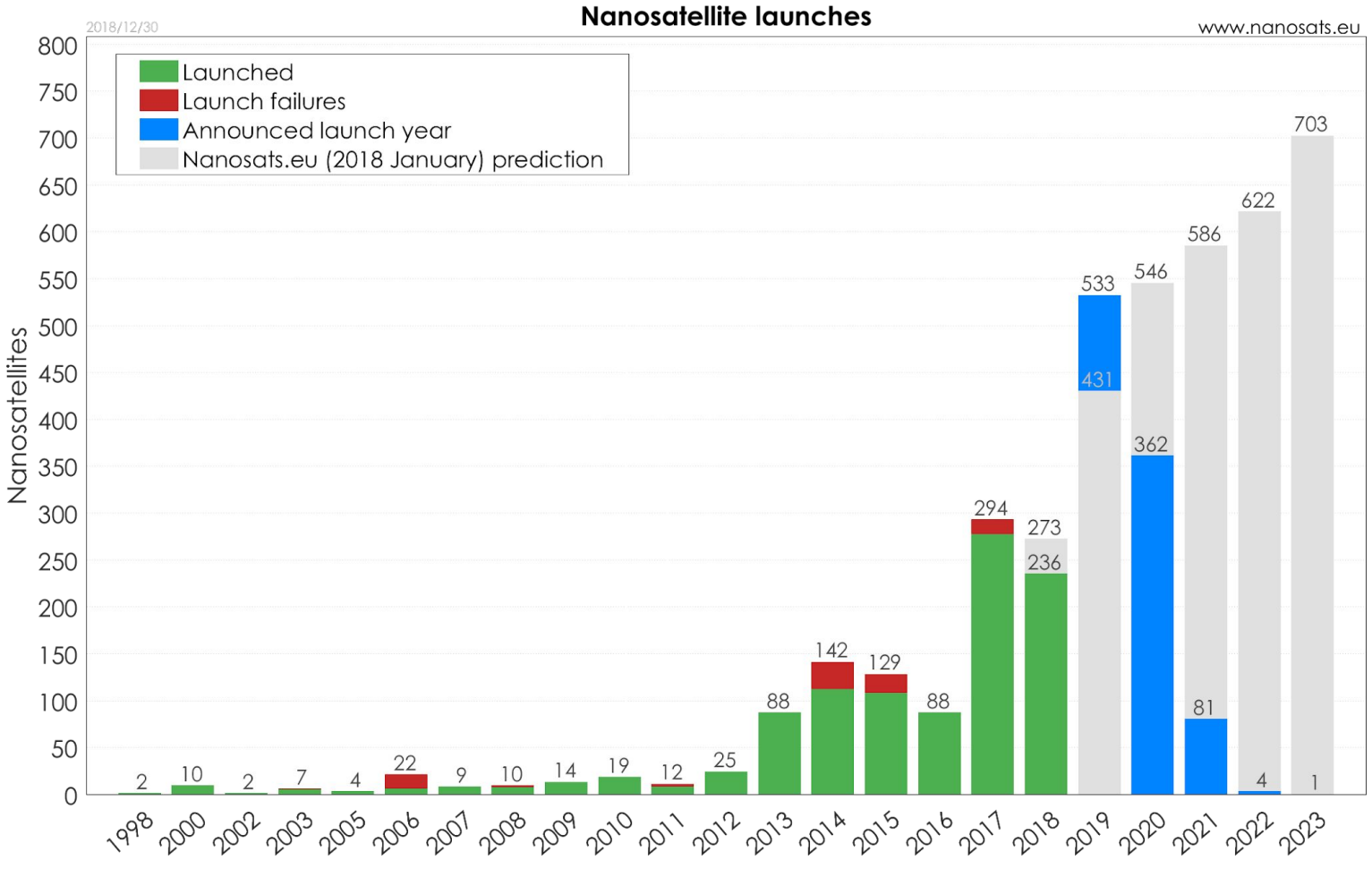


Standardized Form Factors

- Standardized Buses
- Standardized Instruments
- Commercial off-the-shelf is a legitimate option

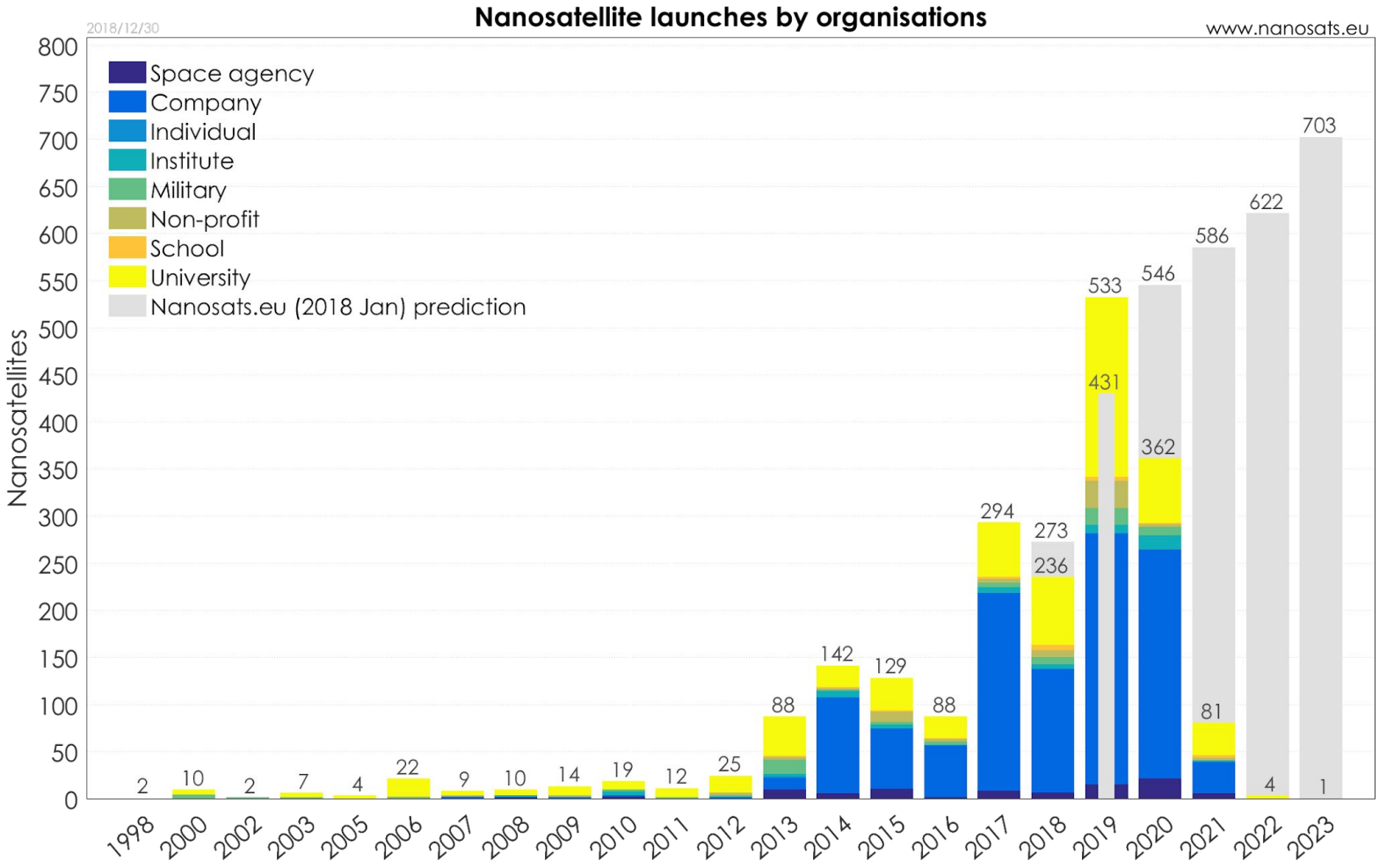
The barrier of entry is dramatically reduced.

Launches



nanosats.eu/index.html#figures

Launches by Institutions



nanosats.eu/index.html#figures

Space Debris

- Hundreds of new CubeSats deployed each year
- Trending toward a 50 percent increase in the number of collisions
- Over the past 25 years, the number of tracked space objects >4in has increased from 8k to 18k.
- Only 7% of these objects are operational satellites.
- Each of the LEO satellites operated by ESA has to perform a maneuver to avoid a tracked piece of debris at least once a year

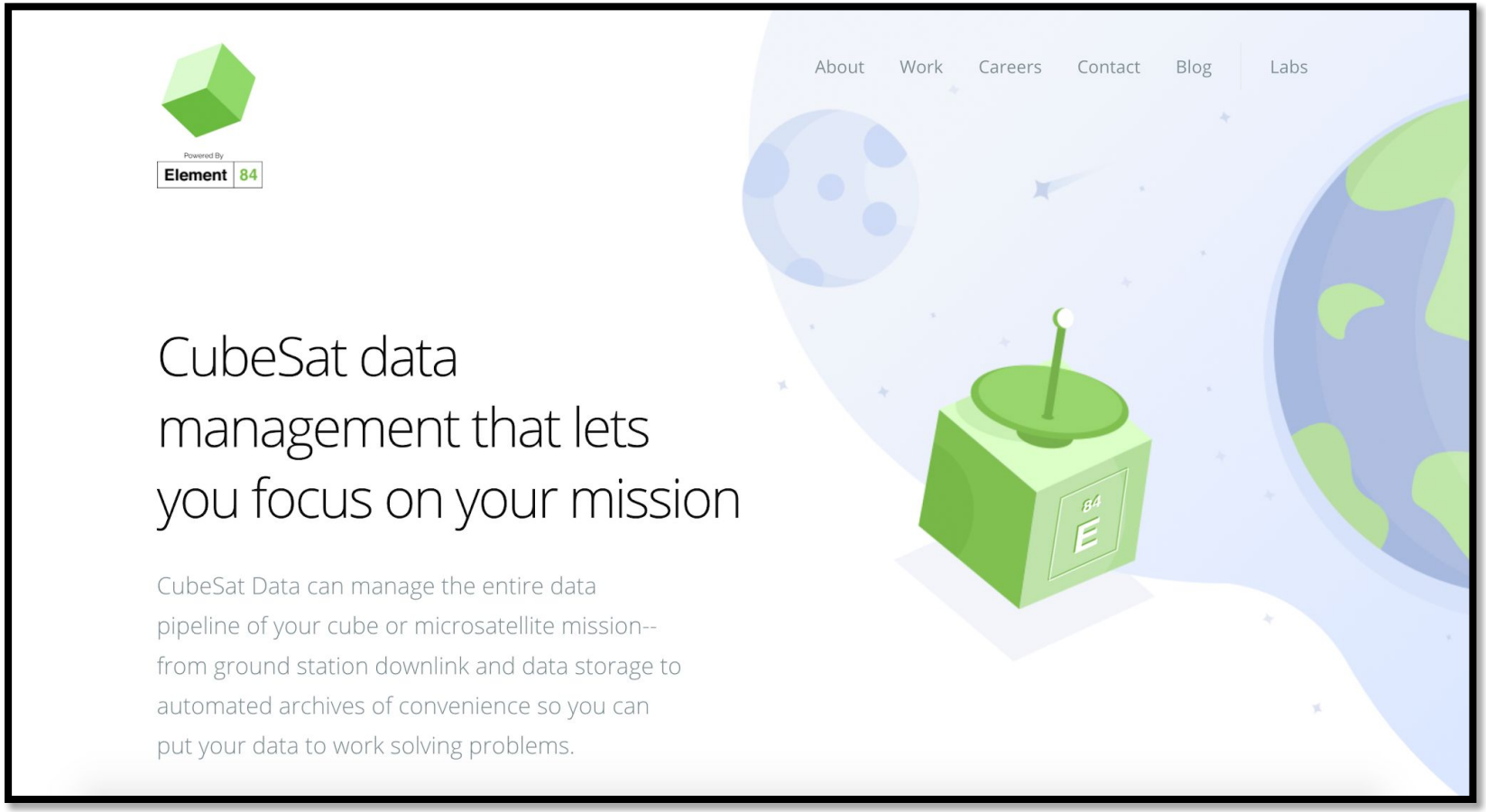


en.wikipedia.org/wiki/Sentinel-1
https://www.space.com/36506-cubesats-space-junk-apocalypse.html

More, increasingly diverse satellites have more, increasingly diverse data needs.

Organizations investing in CubeSats want to focus on their missions, rather than their data management pipelines.

cubesatdata.com



FilmDrop

Satellite Data Pipeline Processing and Management



Archive and Storage



Data is stored and indexed with mission-optimized, open data and metadata standards.

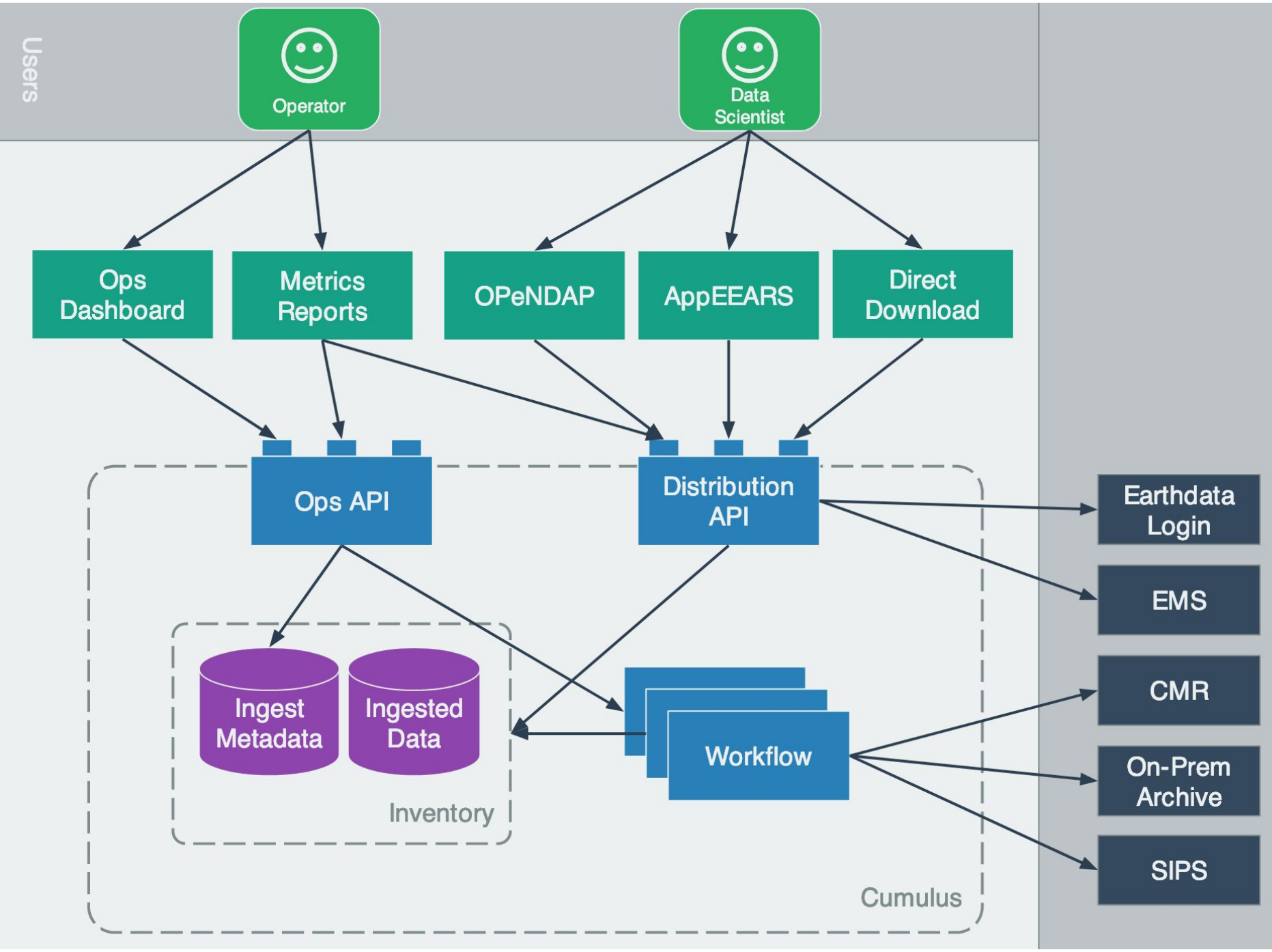
Processing Pipelines

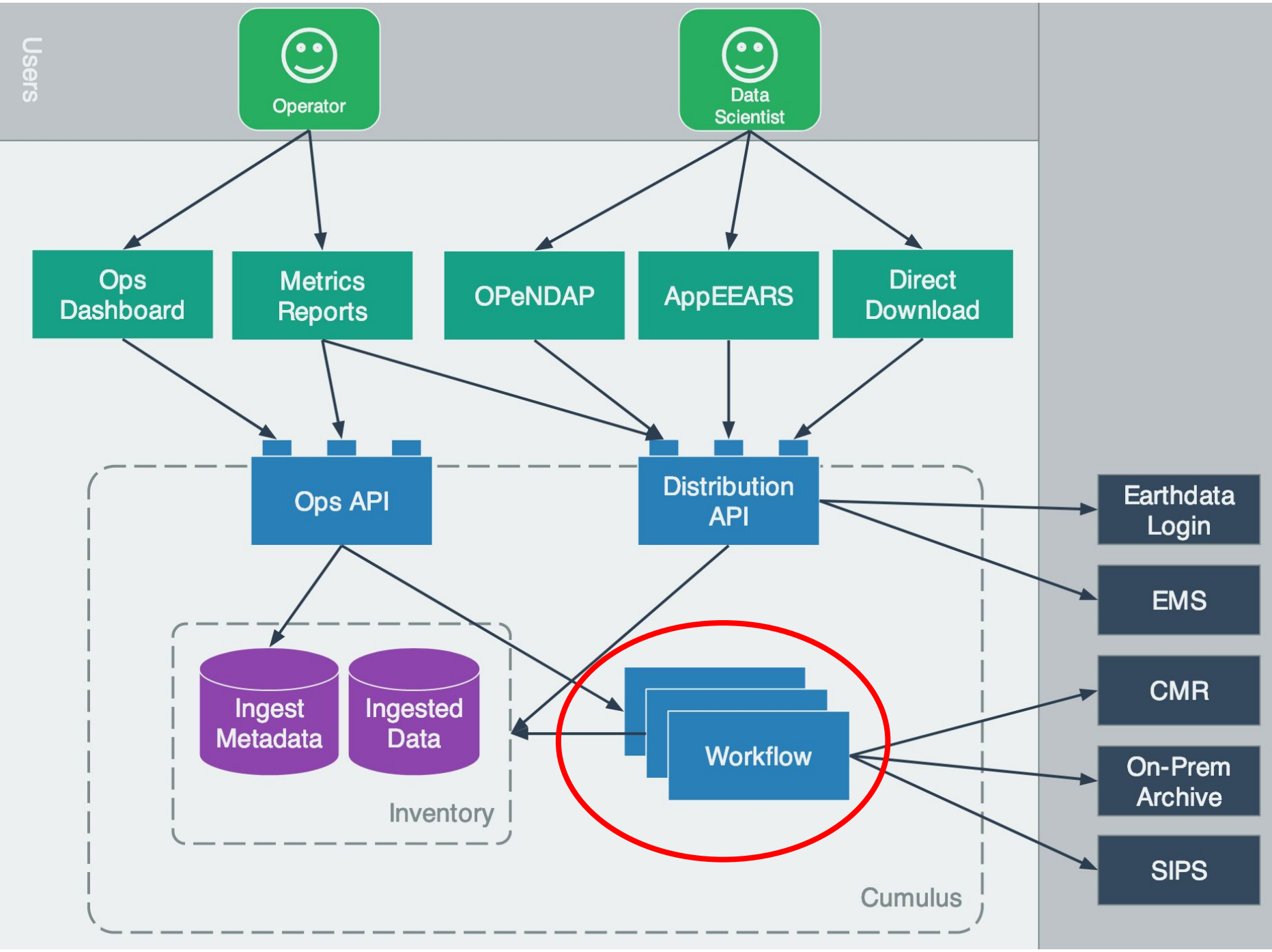
Only write processing functions, FilmDrop deploys and scales



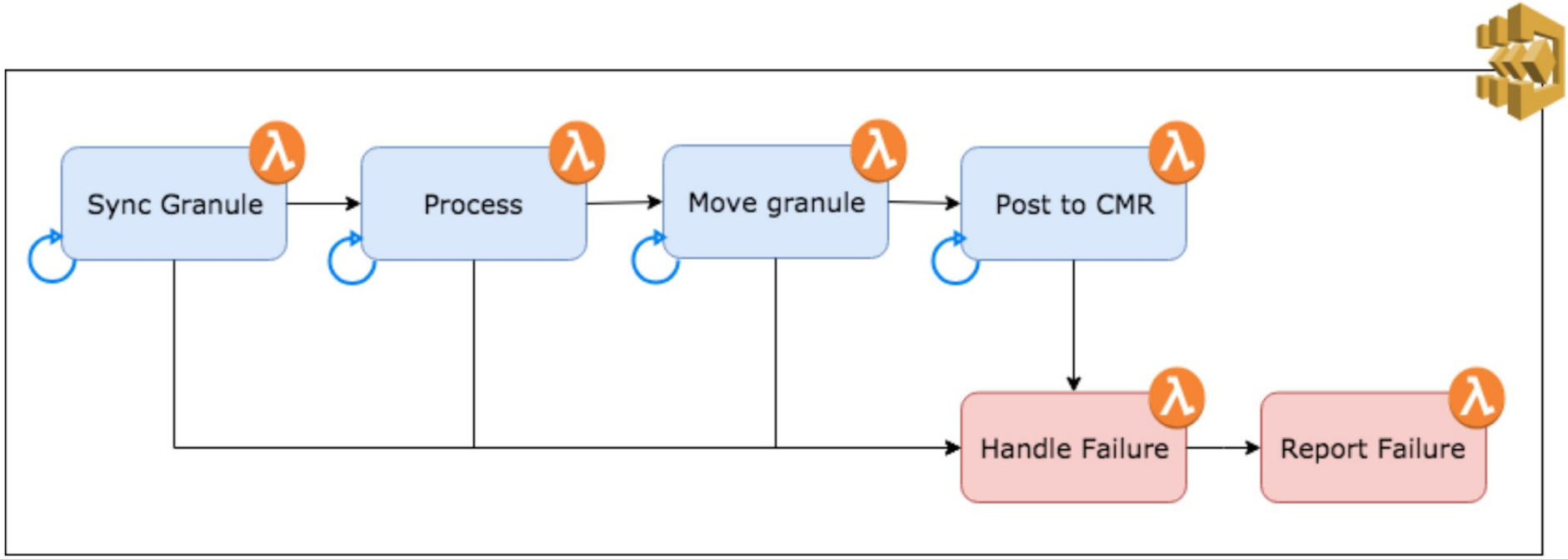
Cumulus

- NASA EOSDIS' cloud-native data processing and distribution framework
- https://github.com/nasa/cumulus
- https://github.com/nasa/cumulus-dashboard


Users
Operator
Data Scientist
Ops Dashboard
Metrics Reports
OPeNDAP
AppEEARS
Direct Download
Ops API
Distribution API
Earthdata Login
EMS
CMR
On-Prem Archive
SIPS
Ingest Metadata
Ingested Data
Inventory
Workflow
Cumulus


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Ops API
Distribution API
Earthdata Login
EMS
CMR
On-Prem Archive
SIPS
Ingest Metadata
Ingested Data
Inventory
Workflow
Cumulus

Cumulus


Sync Granule
Process
Move granule
Post to CMR
Handle Failure
Report Failure

Cumulus

- Robust workflow scaling
- Edge-case, failure handling
- Serverless (Lambda) or Clusters
- Science developers write science code
  - We build and operate the pipelines

Discovery and Access

Search, discover, and visualize your data at scale and make it available via services for your team or customers.



sat-api

https://github.com/sat-utils/sat-api

- STAC dynamic API reference implementation
- Node library for a (STAC) API
- Deploy your own API: https://github.com/sat-utils/sat-api-deployment
- Crawl existing static STAC catalogs to index
- Keep up to date by subscribing to STAC SNS topics

SAT-API Browser

D QUERY | RESULTS | CANDIDATES

Date Range

Start Date (required) 2019/07/14 06:33 PM

End Date (required) 2019/07/15 06:33 PM

Area Of Interest (required)

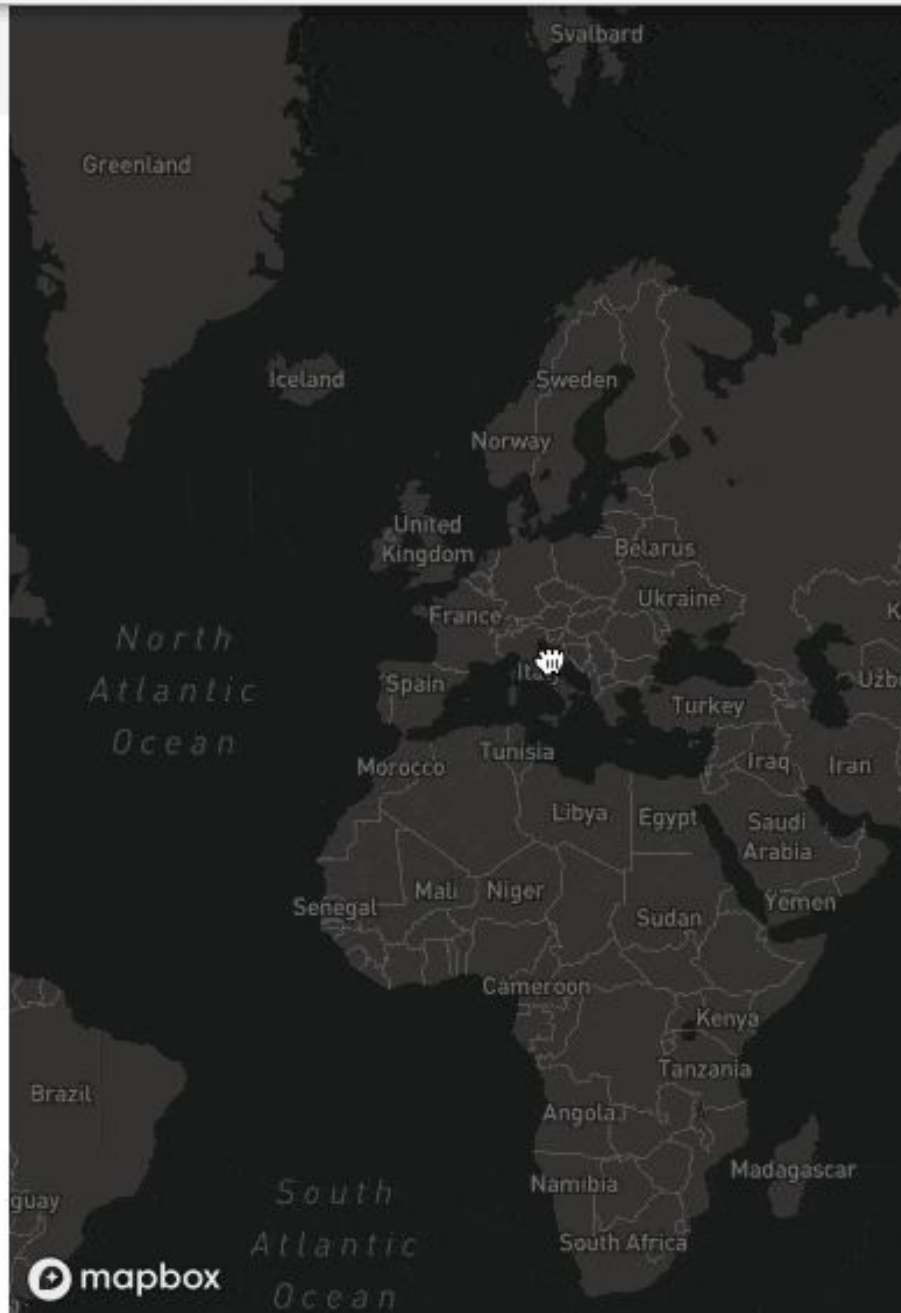
DRAW BBOX ON MAP

Property Filters

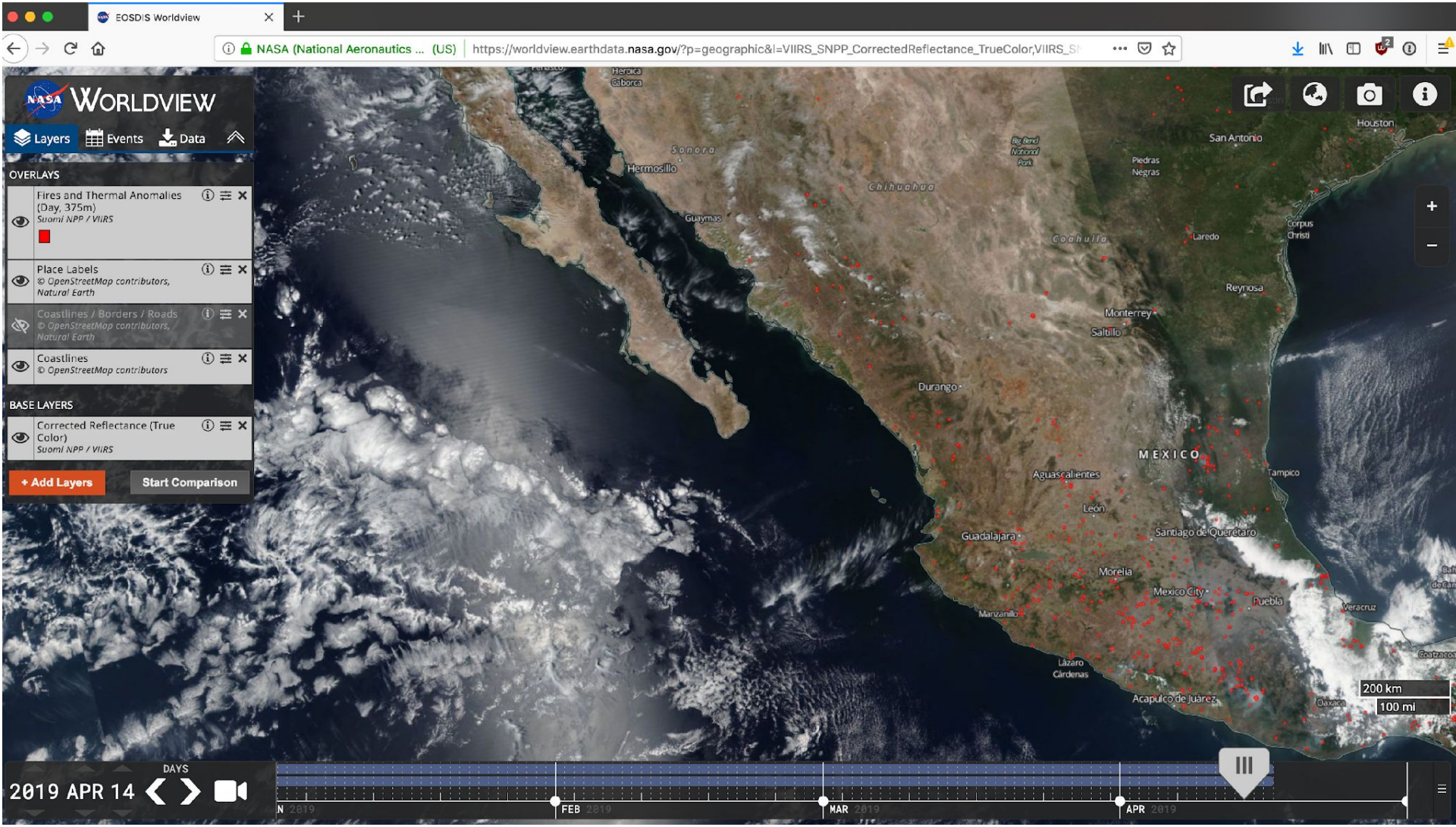
Add new filter

Select a property to filter

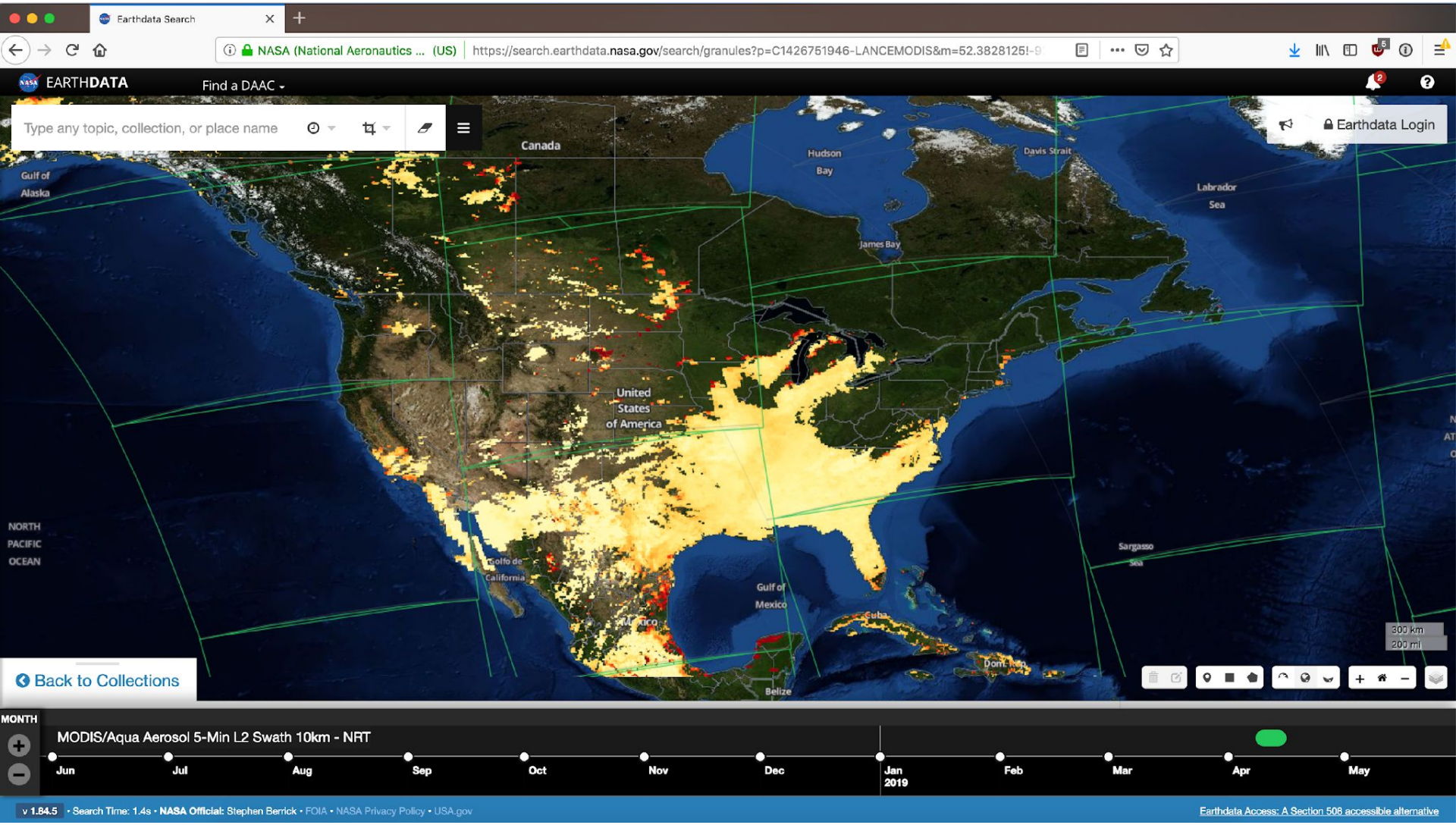
SUBMIT QUERY



OnEarth


EOSDIS Worldview
NASA (National Aeronautics ... (US)
https://worldview.earthdata.nasa.gov/?p=geographic&l=VIIRS_SNPP_CorrectedReflectance_TrueColor,VIIRS_S
WORLDVIEW
Layers
Events
Data
OVERLAYS
Fires and Thermal Anomalies (Day, 375m)
Suomi NPP / VIIRS
Place Labels
© OpenStreetMap contributors, Natural Earth
Coastlines / Borders / Roads
© OpenStreetMap contributors, Natural Earth
Coastlines
© OpenStreetMap contributors
BASE LAYERS
Corrected Reflectance (True Color)
Suomi NPP / VIIRS
+ Add Layers
Start Comparison
Hermosillo
Guaymas
Chihuahua
Coahuila
Durango
Monterrey
Saltillo
Reynosa
Laredo
Corpus Christi
San Antonio
Houston
Piedras Negras
Aguascalientes
León
Guadalajara
Santiago de Querétaro
Morelia
Mexico City
Puebla
Veracruz
Tampico
Manzanillo
Lázaro Cárdenas
Acapulco de Juárez
MEXICO
200 km
100 mi
DAYS
2019 APR 14
FEB 2019
MAR 2019
APR 2019

Earthdata Search Client



JupyterHub

- Shared clusters for running single-user Jupyter notebooks
- Users run their workloads where the data lives
- Easy to bundle and share analysis artifacts



FilmDrop

Satellite Data Pipeline Processing and Management



Free and open source software is empowering SmallSat operators to focus on their missions.

Open standards are empowering users to build out meaningful interactions with SmallSat data.

Thank you

Matthew Hanson

@GeoSkeptic

Element84.com

@element84

Element 84